

Mathematical Summary of A* Search for Quantized MLP Optimization

Analysis of Parameter Update Strategies and Cost Functions

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Abstract

This document provides a mathematical summary of the A* search algorithm components—specifically the cost functions and parameter update strategies—used to optimize a Quantized Multi-Layer Perceptron (MLP). The optimization seeks to find an optimal set of discrete, quantized weights by navigating a finite search space.

1 A* Search Cost Components

The A* algorithm evaluates nodes n in the search space using a total estimated cost function $f(n)$, which is typically a combination of the cost-to-come $g(n)$ and the heuristic cost-to-go $h(n)$.

1.1 Heuristic Cost (Cost-to-Go), $h(n)$

The heuristic $h(n)$ represents the estimated cost from the current state (node n) to the goal state. In this context, $h(n)$ is defined as the **model's loss** on the training data (X, Y) :

$$h(n) = \text{Loss}(\text{MLP}_n(X), Y)$$

where MLP_n is the Quantized MLP model associated with node n , and Loss is the chosen loss function (e.g., Mean Squared Error).

1.2 Cost-to-Come, $g(n)$

The cost-to-come $g(n)$ is the accumulated actual cost from the initial state n_0 to the current state n . For a transition from a parent node n_{parent} to a neighbor node n , the cost is updated by a step cost Δg :

$$g(n) = g(n_{\text{parent}}) + \Delta g(n_{\text{parent}} \rightarrow n)$$

The initial cost is $g(n_0) = g_{\text{ini}}$, where $g_{\text{ini}} = \text{self.g_initial_value}$. The step cost Δg is dependent on the chosen update strategy.

1.3 Total Estimated Cost, $f(n)$

The total estimated cost $f(n)$ is calculated based on $g(n)$ and $h(n)$, and may incorporate a scaling factor to dynamically adjust the influence of g versus h . Let $T_L = \text{self.target_loss}$ be the target loss.

1.3.1 Standard A* or Unscaled f

If $\text{self.update_strategy} = 0$, or if $\text{self.scale_f} = \text{False}$ (for strategies 1 and 2), the standard A* function is used:

$$f(n) = g(n) + h(n)$$

1.3.2 Scaled f (for Strategies 1 and 2 with $\text{self.scale_f} = \text{True}$)

If a scaling factor is applied, the cost-to-come $g(n)$ is scaled by a term that depends on the current loss $h(n)$ relative to the target loss T_L . This essentially prioritizes nodes with a loss close to the target.

$$f(n) = \max\left(0, 1 - \frac{T_L}{h(n)}\right) g(n) + h(n)$$

2 Update Strategies for Step Cost Δg

The step cost $\Delta g(n_{parent} \rightarrow n)$ for moving from a parent node n_{parent} to a neighbor node n depends on the parameter `self.update_strategy`.

Let $h_{parent} = h(n_{parent})$ and $h = h(n)$. Let h_0 be the loss of the initial node.

- $C = \text{initial_node.quantized_mlp.possible_confgurations}$: Total number of possible quantized weight configurations.
- $M = \text{self.max_iterations}$: Maximum number of search iterations.
- $\alpha = \text{self.alpha}$: Mixing factor ($0 \leq \alpha \leq 1$).
- $g_{step} = \text{self.g_step}$: A constant step value.
- $k = \text{iteration} + 1$: Current iteration index.

2.1 Strategy 0: Constant Step Size

The step cost is a fixed parameter:

$$\Delta g = g_{step}$$

2.2 Strategy 1: Loss-Based Scaling (Log-Scale Configuration)

The step cost is scaled based on the relative change in loss from the parent node and the initial node. Note that $\log(C)$ implies the natural logarithm.

$$\Delta g = \frac{1}{\ln(C)} \left(\alpha \frac{h}{h_{parent}} + (1 - \alpha) \frac{h}{h_0} \right)$$

2.3 Strategy 2: Normalized Constant Step Size

The step cost is a small, constant fraction of the maximum allowed iterations:

$$\Delta g = \frac{1}{M}$$

2.4 Strategy 3: Loss-Based Scaling with Iteration-Dependent Exponent

This strategy uses a complex scaling factor that is further modulated by the iteration number k using a base-10 logarithm.

$$\Delta g = \frac{1}{M \cdot \log_{10}(C)} \left(\alpha \frac{h}{h_{parent}} + (1 - \alpha) \frac{h}{h_0} \right)^{\max(1, \log_{10}(k))}$$

3 Quantization and State Space

The search space is discrete, defined by the quantization of the model parameters.

3.1 Quantization Step

The quantization step δ is determined by the quantization factor $Q = \text{self.quantization_factor}$:

$$\delta = \frac{1}{Q}$$

A weight w is quantized to w_q by rounding to the nearest multiple of δ :

$$w_q = \frac{\text{round}(w \cdot Q)}{Q}$$

3.2 Total Possible Configurations

The total number of possible quantized configurations, C , is used to normalize the step cost in Strategies 1 and 3. Let P be the total number of scalar parameters and $r_{\max} = \text{self.parameter_range}[1]$ be the maximum absolute parameter value (assuming a symmetric range). The number of possible values for a single parameter is $2 \cdot r_{\max} \cdot Q + 1$.

$$C = (2 \cdot r_{\max} \cdot Q + 1)^P$$